Jev for Scientific Decisions: Evaluating Semantic Choices and Their Consequences

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***Abstract*—Scientific workflows often require choosing among known relations before a deterministic calculation can proceed. Whether observations share a culture, treatment or reference standard can change the scientific meaning of the resulting count or comparison. Once the evidence is prepared and the calculation is written as code, this choice reduces to one semantic judgment among listed candidates. We evaluate Jev on this step, using a harness that follows its documented guidance. The study compares twelve model configurations on twenty Choices in ten scientific cases built from published sources, each repeated five times. We measure semantic selections, downstream outputs and final claim labels separately. Jev tied with five other configurations at complete semantic correctness and achieved the lowest observed median latency and known mean cost among successful responses. On one question, whether samples shared a culture, three comparison models made seven wrong selections that changed downstream counts while preserving the correct final label. These results support using Jev to select relations when the evidence and candidates are prepared in advance, and show that evaluating it requires checking the relations and quantities that a workflow will reuse.**

Code: `scientific-decision-eval`

***Index Terms*—downstream correctness, Large language model evaluation, Scientific workflows, Semantic correctness, Semantic reasoning**

I. Introduction

Scientific workflows often turn on small decisions about what an observation represents. Ten measurements may describe independently grown cultures or repeated samples from a single culture. That distinction determines what a program should count and which claims the evidence supports. Similar decisions arise when identifying shared calibration standards, paired simulations or experimental assignments. Such decisions have a short list of possible answers. Once the candidates are listed and the calculation is written as code, what remains for the model is one bounded selection that depends on reading scientific language correctly.

Jev answers questions about supplied information and returns structured results [1]. Its Choice interface selects from explicit candidates [2]. The documented division of labor assigns numerical operations to code and semantic judgments to the model [3]. For these semantic judgments, the practical questions are whether a model makes them correctly at low cost and latency, whether doing so requires additional reasoning, and whether evaluation captures their downstream consequences.

Scientific question answering and claim verification already provide relevant evaluation settings. PubMedQA pairs research questions with abstract context [4], and Jevals uses a binary version to evaluate Jev [5]. SciFact evaluates scientific claims together with supporting evidence [6]. More recently, Ho et al. [7] showed that correct scientific claim labels can coexist with incorrect cell-level rationales. Sprague et al. [8] found that chain-of-thought prompting helps mainly on mathematical and symbolic tasks, and PAL [9] delegates such steps to an interpreter. Our harness assigns these operations to fixed code. We examine how selected relations determine subsequent calculations and whether correct claim labels are accompanied by correct scientific outputs.

We contribute scientific decision cases built from published sources, a shared harness and a comparison of Jev with eleven other configurations. Our analysis connects semantic correctness to derived outputs and final labels, and compares request latency, response cost and reported reasoning tokens.

II. Scientific Decisions and Evaluation

A. Cases and the shared harness

The model comparison uses twenty scientific Choices in ten cases, with two questions per case. The questions ask what counts as one experimental unit, whether units share a history such as a common culture, whether observations depend on one another, and which protocol applies to which observations. Each case also has a claim that we constructed about the source study. The claim consists of clauses, and shared code judges each clause from the selected relations and derived quantities. Sources span manufacturing and materials, physical simulations, climate, metrology and biological experiments. These cases belong to a development set of forty scientific Choices in twenty case groups. We call the twenty Choices answered by the models *model-resolved* and the other twenty *code-resolved*. Deterministic parsers written for the prepared passages answer the code-resolved Choices without a model call. A parser leaves its Choice unresolved unless the passage matches the expected wording in full. Appendix A lists the cases and their reference interpretations.

Each model receives the same prepared evidence passages, structured records, question instructions and candidate relations. These inputs adapt or transcribe the source documents. Reference answers and the constructed claims are withheld from the models.

Following Jev's guidance, the harness asks atomic questions, points each question to the supplied evidence and keeps counting, filtering, arithmetic and label composition in code. Jev uses its native Choice interface; the other models return candidate identifiers through a constrained JSON schema. Both questions are submitted together.

B. Three levels of correctness

Let e denote the supplied evidence and records, and Q the questions with their candidate relations. The model f selects a relation for each Choice, yielding $\hat{\mathbf{r}}$. The shared deterministic program g applies the case's computation and claim-assessment rules to produce scientific outputs $\hat{\mathbf{z}}$ and a final label $\hat{y}$:

$$\hat{\mathbf{r}} = f(e, Q), \qquad (\hat{\mathbf{z}}, \hat{y}) = g(e, \hat{\mathbf{r}}). \tag{1}$$

Figure 1 shows this division. The reference relations, outputs and label are $\mathbf{r}^*$, $\mathbf{z}^*$ and y^*.

Semantic correctness requires the selected candidate for Choice k to match its reference exactly: $\hat{r}_k = r_k^*$. **Downstream exactness** requires agreement on every output field that the case definition declares (Appendix Table V), including counts, Boolean relations, symbolic expressions and unknown values. These fields exclude clause judgments and labels. **Label correctness** requires $\hat{y} = y^*$. The program g combines the clause judgments into the final label: any contradicted clause makes it `inconsistent`; otherwise any unresolved clause makes it `insufficient_evidence`; a claim whose clauses are all supported is `consistent`. We also report **joint correctness**, requiring all semantic selections and downstream outputs to agree. Appendix B specifies the output fields and scoring procedure.

These checks answer different questions. The program in Equation 1 can map distinct relations to the same quantities or final label. It can also execute a wrong relation exactly and produce a wrong scientific result. The reference label of all ten model-resolved cases is `inconsistent`, so an always-`inconsistent` baseline solves the label task. Semantic and joint correctness provide the substantive quality comparison.

C. Experimental comparison

We compared twelve configurations on all ten cases, each repeated five times. Each configuration therefore had fifty planned requests and one hundred semantic answers. Configurations were evaluated in three groups, with serial requests, shuffled case order and rotating model order within each group. Repetitions measure variation on the same inputs and do not enlarge the scientific sample, which remains ten cases containing twenty Choices. Two checks require no model calls. A code-only run answers the code-resolved Choices and leaves the model-resolved ones unresolved. A reference-oracle run passes the reference answers for all forty Choices to the shared code, confirming that the derivation code reproduces every reference output and label.

The primary scores divide by all planned requests or answers, so a missing response earns no credit. We record missing responses separately from wrong selections. We also report correctness conditional on receiving an answer (Appendix C). Each request was sent once, with no fallback to another hosting provider. We requested medium reasoning effort for the OpenAI and Qwen configurations, and high effort for DeepSeek and Kimi. Anthropic configurations used adaptive thinking with medium effort. The eleven comparison configurations had an output limit of 32,768 tokens; Jev requests left the reasoning setting and output limit unspecified. Exact model identifiers and service configurations are provided with the released code.

Latency measures elapsed client request time. We report its median and nearest-rank 95th percentile among successful responses. Cost is the known mean charge per successful response, using provider-reported charges or estimates from token usage and the applicable prices.

III. Results

A. Semantic and downstream correctness

Jev answered all one hundred planned semantic questions correctly and obtained all fifty downstream results (Table I). GPT-5.6 Sol [10], GPT-6 Astra [11], Claude Sonnet 5 [12], Claude Opus 5 [13] and Kimi K3 [14] achieved the same scores. Luna made one wrong selection, Qwen Flash [15] one and Qwen Max [16] five. All seven concerned the same culture question.

Several configurations returned fewer than fifty responses. Terra and DeepSeek configurations [17], [18] answered every received question correctly, but missing responses reduced their primary scores. Conditional scores in Appendix C separate these missing responses from semantic errors.

B. Correct labels can conceal wrong quantities

The culture case uses the experiments of Luria and Delbrück [19]. Ten samples in Experiment 11a come from one culture; ten in Experiment 11 come from separately grown cultures. The two series therefore contain one and ten independent cultures, written $(1, 10)$. The question asks whether the samples within each series come from one culture or from independent cultures.

Table II shows the wrong selections across five repetitions of the culture question. Luna and Qwen Flash each erred once. Qwen Max selected three different wrong relations across its five repetitions. In all seven responses, the second Choice correctly identified colonies as clonal descendants. That answer contradicted the clause stating that each colony represented an independent mutation origin, so the final label remained `inconsistent` despite the wrong counts of independent cultures.

Across the 588 received results, all final labels were correct, while 581 downstream outputs were exact. This difference represents seven responses on one scientific question. It exposes a concrete failure that a label-only comparison would miss: the workflow can return the expected verdict while passing an incorrect quantity to subsequent analysis.

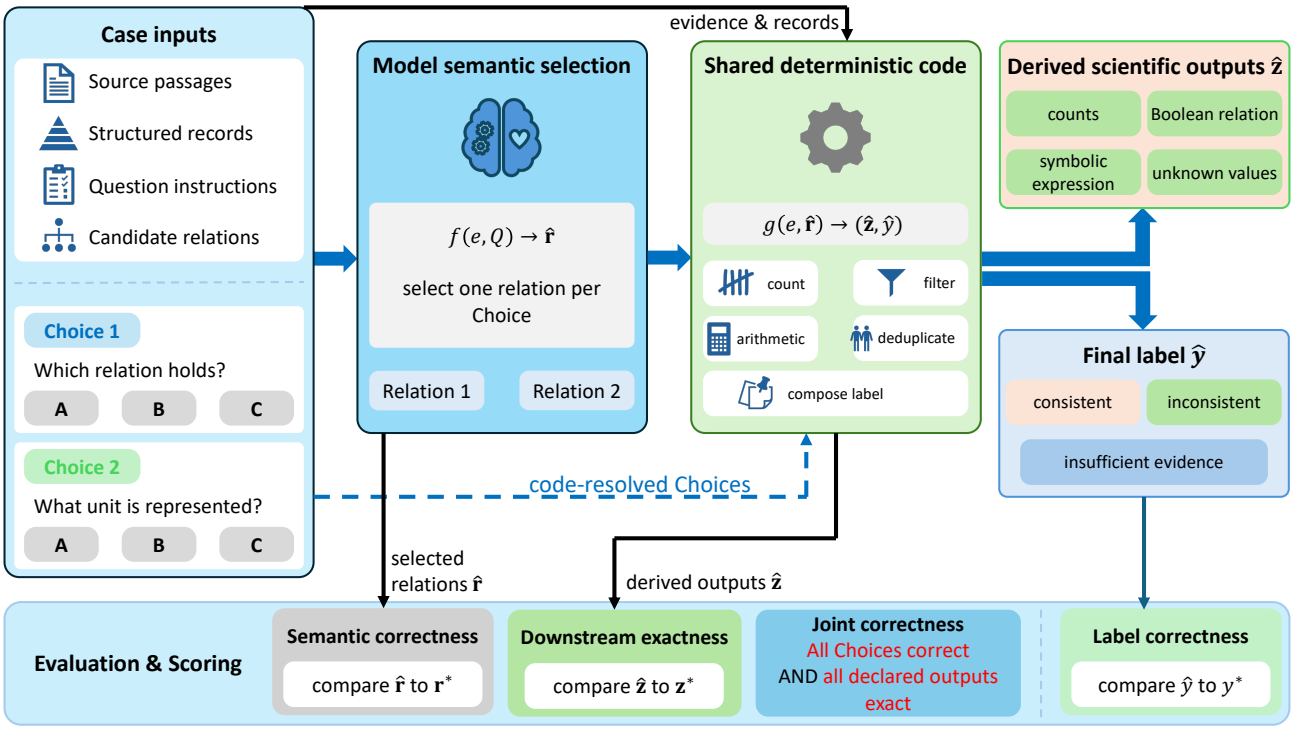

Fig. 1. The model selects relations from supplied evidence; shared code derives scientific outputs and composes a claim label. We score the selected relations, derived outputs and final label against separate references. Code-resolved Choices bypass the model.

C. Reasoning tokens, latency and cost

Reported reasoning tokens did not track correctness. Sol, Astra, Sonnet and Opus reported almost no reasoning tokens and answered every Choice correctly. On the culture question, DeepSeek V4 Pro used a number of reasoning tokens similar to Qwen Max's and answered all five repetitions correctly; Qwen Max erred in all five. The configurations that reported the most reasoning tokens were also the slowest, with median latencies of 5.4 to 14.2 s.

Jev had the lowest observed successful-request median latency, 0.335 s, with a p95 of 0.442 s. Its known mean cost was approximately $0.000060 per successful response. Among the configurations that returned all fifty responses and answered every Choice correctly, Sol had both the next lowest median, 1.385 s, and the next lowest cost, $0.003047 per successful response. In this comparison, Jev matched these configurations in correctness while achieving lower median latency and lower known mean cost among successful responses.

IV. Implications for Scientific Workflows

The comparison supports using Jev where evidence has been prepared, candidate relations are explicit and downstream operations are specified. Preserving the semantic selections also makes it possible to inspect which relation caused a downstream discrepancy. Because configurations that reported almost no reasoning were fully correct, a workflow can reserve reasoning models for steps that cannot be reduced to a listed choice, such as extracting evidence or proposing candidates.

Ho et al. [7] found that correct claim labels can coexist with incorrect rationales; the culture case shows the same gap for quantities that a scientific workflow could reuse. Because a final label combines several clause judgments, one correctly contradicted clause can conceal an error elsewhere. Evaluations of components that select from fixed candidates should therefore keep both the reference relations and the declared downstream outputs.

V. Conclusion

We compared Jev with eleven other configurations on relation-selection questions with prepared evidence and candidates, using the same deterministic code for every configuration. Jev answered every Choice correctly and was the fastest and least expensive configuration we measured. Extended reasoning was not needed to reach complete correctness. Separating semantic selections, downstream outputs and final labels revealed errors hidden by the final verdict.

Limitations

The comparison uses a development set of twenty model-resolved Choices in ten cases. The same curated English inputs recur across repetitions, and six configurations achieve complete correctness. All observed semantic errors concern one Choice. These results characterize only the supplied questions; evaluating transfer requires cases from experiments not represented in this collection and broader relation coverage.

Every model-resolved case has the reference label `inconsistent`. None of the reference answers indicates

TABLE I

COMPARISON ON TWENTY CHOICES IN TEN CASES, REPEATED FIVE TIMES PER CONFIGURATION. RETURNED COUNTS COMPLETE RESPONSES. CORRECTNESS IS COMPUTED OVER ALL PLANNED REQUESTS, WITH NO CREDIT FOR MISSING RESPONSES. DOWNSTREAM AND JOINT COUNTS COINCIDE IN THESE RESULTS. COST IS THE KNOWN MEAN COST PER 1,000 SUCCESSFUL RESPONSES. THE MARKER r DENOTES REPORTED CHARGES AND e USAGE-BASED ESTIMATES. PREPARATION COSTS AND UNMEASURED CHARGES ARE EXCLUDED. REASONING TOKENS ARE MEDIANS. JEV DOES NOT REPORT THEM.

	Responses and correctness				Successful-request resources			
Model	Returned /50	Semantic /100	Downstream = Joint /50	Label /50	Reasoning tokens	Median (s)	p95 (s)	Cost ($/1,000)
Jev 1.13	50	100	50	50	—	0.335	0.442	0.0595[r]
GPT-5.6 Luna	50	99	49	50	32	1.509	2.753	0.2025[e]
GPT-5.6 Terra	49	98	49	49	0	1.166	2.147	1.5391[e]
GPT-5.6 Sol	50	100	50	50	0	1.385	2.370	3.0470[e]
GPT-6 Astra	50	100	50	50	0	2.384	4.418	7.6174[e]
Claude Sonnet 5	50	100	50	50	0	2.162	5.422	4.2502[e]
Claude Opus 5	50	100	50	50	0	3.237	3.967	10.6965[e]
Qwen3.8 Flash	42	83	41	42	636	14.182	34.320	0.4013[r]
Qwen3.8 Max 0902	50	95	45	50	521	14.236	21.393	4.8302[r]
DeepSeek V4.1 Flash	48	96	48	48	370	5.449	8.821	0.3508[r]
DeepSeek V4 Pro 0813	49	98	49	49	429	7.569	10.086	3.4299[r]
Kimi K3	50	100	50	50	244	11.116	25.744	5.4663[r]

TABLE II

WRONG SELECTIONS ON THE CULTURE QUESTION. "SHARED" MEANS SAMPLES CAME FROM ONE CULTURE; "SEPARATE" MEANS THEY CAME FROM SEPARATELY GROWN CULTURES. INDEPENDENT-CULTURE COUNTS ARE ORDERED AS (11A, 11), WITH REFERENCE VALUES $(1, 10)$. OCCURRENCES COUNT HOW OFTEN EACH WRONG SELECTION APPEARED IN FIVE REPETITIONS PER MODEL.

	Selected answer		Independent	Occurrences	
Model	Experiment 11a	Experiment 11	cultures	out of five	Final label
GPT-5.6 Luna	Separate	Shared	$(10, 1)$	1	`inconsistent`
Qwen3.8 Flash	Separate	Shared	$(10, 1)$	1	`inconsistent`
Qwen3.8 Max 0902	Shared	Shared	$(1, 1)$	2	`inconsistent`
	Separate	Shared	$(10, 1)$	2	`inconsistent`
	Separate	Separate	$(10, 10)$	1	`inconsistent`

insufficient evidence, an answer outside the listed candidates, or conflicting evidence. Assessing label discrimination and abstention requires cases with a mix of reference labels and cases where the evidence is uncertain.

The evaluation begins with supplied evidence passages and structured records. The harness comparison uses one implemented configuration per model. Measuring the effect of each harness choice, such as submitting both questions together, would require a separate ablation study. We did not vary reasoning effort within a model, and six configurations reached complete correctness, so the comparison cannot show whether more reasoning would help on harder cases.

Latency and cost depend on the serving provider, the time of the runs, reasoning settings and cache states. The resource comparison covers successful model requests. Charges for missing responses, data preparation costs and local computation costs are unmeasured.

APPENDIX A
SCIENTIFIC QUESTIONS AND CASE CONSTRUCTION

The collection contains 20 groups, 40 scientific Choices, and 25 claim outcomes. A group shares evidence across related questions and can support more than one claim. Ten groups, comprising 20 scientific Choices, follow deterministic rules developed on the prepared passages. The other ten groups each contain two model-resolved Choices. Tables III and IV list the complete collection and the resulting computations. The 21 source records belong to 18 documented source families. Related documents from the same experiment remain in one family.

A. Evidence and reference construction

Each input supplies a target, an explicit scope, selected evidence passages, and structured records when the calculation requires them. The passages and records were prepared from articles, supplements, scientific project documentation, a methods handbook, and an industrial case report.

For each of the twenty questions answered by the models, we specified a reference answer and a rationale based on the supplied source evidence. One author reviewed these answers and their supporting rationales. Expected scientific outputs were derived from the reference answers and supplied records using the rules described for each case. Reference labels indicate whether the resulting quantities and relations

TABLE III
SCIENTIFIC RELATIONS AND DERIVED QUANTITIES FOR THE TEN GROUPS RESOLVED BY DETERMINISTIC RULES DEVELOPED ON THE PREPARED PASSAGES.

Group	Domain	Relation or unit	Derived scientific quantity	Source
D01	Concrete	Mixing operation and shared prism membership	One mixing operation, three prisms, six half-prisms	[20]
D02	Additive manufacturing	Complete build versus specimens within it	One build for three selected specimens	[21]
D03	Concrete curing	Bath-level control, run boundaries, specimen assignments	Curing-run count remains unknown	[20]
C01	Electroplating	Whole-plot unit for temperature/concentration; strip for current	One whole plot per setting; two strips per current level	[22]
H01	Additive manufacturing	Column-level laser settings versus specimen-level heat treatment	Five selected columns; one heat-treated specimen	[23]
C02	Measurement systems	Wear-scar generation versus repeated scans	Two wear events distinguished by their cutter and wear-scar identifiers	[24]
E001	Semiconductor processing	Crossed implant/anneal memberships; wafer interaction unit	64 setting combinations, one wafer each	[22]
E002	Wind energy	Hourly controller switch and transition exclusion	Minute records do not add allocations; campaign count unknown	[25]
E003	Catalysis	Recovered catalyst carried between reaction cycles	Eight cycles in one documented reuse series	[26]
E008	Chemical process design	Equality of all eight coded factor settings	20 runs, 18 distinct settings, two pure-error degrees of freedom	[27], [28]

TABLE IV
TEN GROUPS IN THE MODEL COMPARISON. EACH MODEL RECEIVES BOTH QUESTIONS FOR A GROUP IN ONE REQUEST. ALL NUMERIC, STRUCTURAL AND LABEL DERIVATIONS USE SHARED CODE. SMBB DENOTES SMOOTHED BIOMASS-BURNING FORCING. CONTRIBUTOR APPEARANCES COUNT EACH ANIMAL ONCE FOR EVERY POOLED LIBRARY TO WHICH IT CONTRIBUTES.

Group	Domain	Two model judgments	Reference downstream result	Source
E004	Powder recycling	Mixed reuse histories; partially known provenance	Exact parent-lot count unknown; assignment incomplete	[29]
E005	Fatigue testing	Outcome-adaptive stress; next test uses a new specimen	Next stress 295 or 355 MPa after failure or runout	[30]
E006	Cosmology	Phase-reversed initial fields; pair-average statistic	100 realizations, 200 executions, 100 ensemble values	[31]
E007	Climate simulation	Two forcing protocols; bundled implementation changes	50 original and 50 SMBB members	[32]
E009	Electrical metrology	Shared reference contribution; bracketing average	Two laboratory results, three visits; covariance retained	[33]
E010	Cloud seeding	Complementary allocations; allocation versus exposure	118 joint assignments, 236 range-level records	[34], [35]
B001	Developmental exposure	Injected F0 dams; maternal, embryonic and germline exposure	162 F3 pups; zero new F3 injections	[36]
B002	Microbial genetics	One culture versus separate cultures; clonal descendants	One versus ten independent cultures; origins unknown	[19]
B003	RNA sequencing	RNA pooled before libraries; aggregate expression profiles	Four libraries, 16 distinct contributors, 32 contributor appearances	[37]
B004	Genome editing	Within-founder mosaicism; inherited phenotype ambiguity	23 founders, 57 reported alleles; validated line count unknown	[38]

support the constructed claims. Quantities not established by the evidence remained unknown.

Reference choices, downstream outputs, and constructed claim labels are withheld from model inputs. Source explanations remain part of the evidence, including explanations that directly establish a relation.

B. Scientific interpretation of the model-resolved cases

The descriptions below give the two judgments, their substantive alternatives, and the reference interpretation for each model group. Questions also provide explicit options for missing evidence, for contradictory evidence and, where applicable, for an answer outside the listed candidates. Missing or ambiguous evidence maps to `not_stated`; unresolved incompatible evidence maps to `conflicting`. An established answer outside the listed alternatives maps to `other`. E004 additionally distinguishes partially known provenance (option `partial_identification`) from complete assignment and absent evidence. The 20 reference answers are substantive options, and the reference outcome of every model-resolved group contradicts at least one clause of its constructed claim.

a) E004: powder reuse histories: From build 8 onward, recovered powder is topped up with previously reused powder, giving the feedstock mixed reuse histories and partial provenance. Alternatives are one common history, virgin re-

placement or separate unmixed histories. The second question asks whether each portion's original lot, mixture fraction and prior-build history are known. The reference is mixed histories with partial identification: a history label does not identify an original lot, so complete assignments and an exact parent-lot count remain unavailable.

b) E005: staircase fatigue testing: The next stress depends on the preceding outcome and is applied to a new specimen. Alternatives are independently randomized, fixed or predetermined stresses, and continued loading of the same specimen or whole-foil tests. Starting at 325 MPa with a 30 MPa step gives 295 MPa after failure and 355 MPa after runout. Replication at a fixed stress remains unknown.

c) E006: paired-fixed cosmological simulations: The paired-fixed subset of the source's N1000 simulation set couples initial fields by phase reversal and contributes pair averages to the ensemble. Alternatives are independent draws, identical fields or output resampling, with individual executions, pair differences or an overall mean as aggregation units. The counts in Table IV describe this coupling and aggregation, not an effective sample size for every evolved observable.

d) E007: climate-ensemble protocol groups: One Choice asks whether the ensemble splits into original and SMBB groups or follows a single protocol throughout. The other asks whether such a split reflects smoothing alone, smoothing bundled with other code changes, or initialization only. The reference is the two-group split with documented emission and carbon-flux code corrections, so the comparison cannot isolate smoothing.

e) E009: correlated calibration uncertainty: The results share reference-calibration uncertainty, and two secondary-laboratory visits form a bracketing average. Alternatives are independent realizations of that calibration or one result copied into both records. The visits could instead be two additional laboratories or a contrast across a deliberate adjustment. The squared uncertainty of the difference contains $-2u_B(\mathrm{std}_{\mathrm{DCV}})^2$, where $u_B(\mathrm{std}_{\mathrm{DCV}})$ is the standard uncertainty of the calibration of the shared 10 V DC voltage standard. Effective sample size remains unknown.

f) E010: paired mountain-range allocation: Complementary allocation makes each decision joint across the two ranges. The alternatives are independent assignments, seeding both or permanent assignment. Allocation establishes neither zero exposure nor confirmed exposure everywhere. The authors' reported totals give $154-27-9 = 118$ joint assignments and 236 range-level records. Monthly generator-exclusion cells sum to eight, whereas the margins and prose report nine. The missing event remains unresolved.

g) B001: administration and developmental exposure: Injection of pregnant F0 dams exposes the mother, embryo, and fetal germline within the source's defined window. The administration alternatives include separate treatment of every generation, F3 animals alone, and isolated germ cells. The exposure alternatives distinguish the full window from maternal-only exposure, mother plus embryo, or exposure of every later descendant generation. The selected dioxin rows contain 162 F3 pups in 14 weaned litters, with zero newly injected F3 animals. Under the source's exposure-window definition, F3 is the first generation beyond that window. The number of distinct F0 ancestors remains unknown.

h) B002: cultures and mutation origins: Section III-B gives the culture relation, and Table II shows its three alternatives. The records cover the complete selected columns of source Tables 1 and 2, with 514 and 624 colonies. Clonal descendants leave the number of mutation origins unknown. Alternatives are one origin per cell, origins created only at selection or repeated images of one colony.

i) B003: pooling and biological attribution: RNA pooling before library preparation produces aggregate profiles. Alternatives are indexed-library pooling, numerical averaging or tissue pooling before extraction. For attribution, they are complete or partial individual profiles or a new animal. Four eight-animal libraries contain 16 distinct contributors and 32 contributor appearances, with zero separately observed contributor profiles. The individual-library arm remains in the evidence but outside the pooled-arm claim.

j) B004: mosaic founders and transmission: Within-founder mosaicism and an inherited albino allele leave the number of validated new germline lines unknown. Alternatives are a uniform allele pair, one founder per allele or no detected edit. For transmission, they are a unique new-allele marker, absent germ cells or verified allele-specific transmission. For founder 30, the author manuscript reports five alleles, whereas published Figure 3C lists six entries. Either count establishes more than two alleles, so the reference is mosaicism with founder 30's exact total unresolved.

C. Deterministic derivation and claim composition

The shared code performs arithmetic, filtering, deduplication, covariance substitution and claim composition. Missing answers and invalid options remain response failures. The code never substitutes zero for an unknown scientific quantity.

For E001–E010 and B001–B004, `conflicting` takes precedence over `other`, then `not_stated`. These alternatives remain distinct and stop label composition. Substantive answers such as E004's partial provenance can produce unknown output fields. For biological compound claims, an inconsistent clause determines the final label even when another clause lacks evidence.

The code-resolved E008 compares complete coded settings. Differing physical-temperature descriptions in its source do not alter this equality calculation.

APPENDIX B
METRIC DEFINITIONS AND OUTPUT COVERAGE

Let $i \in \{1, \ldots, 10\}$ index cases, $t \in \{1, \ldots, 5\}$ repetitions and $k \in \{1, 2\}$ Choices. Write $s_{itk} = 1$ when the received option exactly matches its reference, and zero otherwise. Let d_{it} indicate exact agreement on the case's declared downstream fields, and ℓ_{it} exact agreement on its final label. Missing

TABLE V
DECLARED SCIENTIFIC OUTPUTS USED FOR DOWNSTREAM EXACTNESS. BECAUSE THESE FIELDS CAPTURE ONLY SELECTED CONSEQUENCES OF A CHOICE, SEMANTIC SCORING COVERS DISTINCTIONS THAT DO NOT CHANGE THE RECORDED OUTPUTS.

Case	Declared downstream fields	Coverage of semantic distinctions
E004	Exact parent-lot count; availability of complete lot assignments.	The powder-history relation affects a clause rather than these fields.
E005	Next stress after initial failure; next stress after initial runout; observed replication at a fixed stress.	Specimen continuity is not used by the recorded stress calculation.
E006	Paired realizations; simulation executions; values entering the ensemble.	The phase relation and the averaging operation are not recorded as output fields.
E007	Original-protocol members; revised-protocol members; total members; whether the contrast changes smoothing only.	Different descriptions of a bundled change can share the same Boolean output.
E009	Compared laboratory results; calibration visits; shared covariance term; effective sample size (unknown).	A bracketing average and an adjustment contrast can have the same recorded counts.
E010	Original and retained experimental units; joint assignment events; range-level records; whether assignments to the two ranges are independent.	The exposure interpretation affects a clause rather than these fields.
B001	F3 pups; weaned litters; newly injected F3 animals; first generation beyond the original exposure window; distinct F0 ancestors.	Some alternative dosing descriptions imply the same newly injected F3 count.
B002	Plating counts, independent-culture counts and colony totals by series; exact mutation origins.	The colony–origin relation affects a clause, while the exact origin count stays unknown.
B003	Selected libraries; distinct contributors, overall and by genotype; contributor appearances; separately observed contributor profiles.	Pooling stage affects a clause rather than these fields.
B004	Deep-sequenced hybrid founders; reported mutant alleles; validated new germline lines.	The substantive mosaicism and marker alternatives affect clauses rather than these fields.

responses set each indicator to zero. Over all planned requests, the metrics are

$$A_S = \frac{1}{100}\sum_{i,t,k} s_{itk}, \qquad A_D = \frac{1}{50}\sum_{i,t} d_{it}, \tag{2}$$

$$A_L = \frac{1}{50}\sum_{i,t} \ell_{it}, \qquad A_J = \frac{1}{50}\sum_{i,t} d_{it}\prod_{k} s_{itk}. \tag{3}$$

Downstream exactness requires agreement in value and type for every field in Table V. An omitted value differs from an explicitly unknown quantity, and a Boolean differs from a number. Selected options and final labels are scored separately.

E007 stores its reference label as `inconsistent_in_both_clauses`, which we report as `inconsistent`.

Different semantic selections can produce the same declared output. In the fatigue case, one question asks whether a new specimen receives the next stress setting. Its answer does not enter the recorded stress calculation. Semantic and joint correctness still require the reference selection. Downstream exactness therefore applies to the declared fields rather than every possible consequence of a relation.

APPENDIX C
CORRECTNESS CONDITIONAL ON A RECEIVED RESPONSE

For a configuration with R complete responses, conditional correctness uses the numerators in Table I with denominators $2R$ for semantic answers and R for downstream, joint and label scores. For example, Qwen3.8 Flash scores $83/84$, $41/42$ and $42/42$ for semantic, downstream/joint and label correctness. Missing responses earn no primary-score credit but are excluded from conditional scores: they are neither semantic mistakes nor `not_stated` selections.

Across received responses, 1,169 of 1,176 Choices are correct, 581 of 588 downstream results are exact, and all 588 labels are correct.

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